FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – XI
PAPER –1
TEST DATE: 19-05-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, B, D

Sol. $T_m = \frac{0.5+0.56+0.57+0.54+0.59}{5} = 0.55 \text{ sec}$

$\Delta T_m = 0.03$

$\frac{\Delta T_m}{T_m} = 5.45\%$

2. A, B, D

Sol. At equation

$kx_0 = \frac{A\varepsilon_0 v_0^2}{2d^2}$

Now, small shift 'x' has been given them

$F_{restoring} = \frac{A\varepsilon_0 v_0^2}{2(d+x)} - k(x_0 - x)$

$T = 2\pi\sqrt{\frac{Md^3}{kd^3 - A\varepsilon_0 v_0^2}}$

3. A, C

Sol. Increase in length of A is Δx_1

Increase in length of B is Δx_2

Compression in spring $\Delta x = \Delta x_1 + \Delta x_2$

Spring force = $k\Delta x$

Strain in rod A = $\frac{L\alpha_A T_0 - \Delta x_1}{L}$

Strain in rod B = $\frac{L\alpha_B T_0 - \Delta x_2}{L}$

Force on A $\rightarrow F_A = \dfrac{Y_A A\left[L\alpha_A T_0 - \Delta x_1\right]}{L}$

$F_B = \dfrac{Y_B A\left[L\alpha_B T_0 - \Delta x_2\right]}{L}$

$F_A = F_B = k\Delta x$ (equal stress)

$\Delta x = \dfrac{LT_0(\alpha_A + \alpha_B)}{\left[1 + \dfrac{kL}{A}\left(\dfrac{1}{Y_A} + \dfrac{1}{Y_B}\right)\right]}$

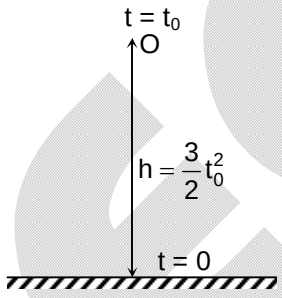
4. D

Sol. Given $\rightarrow \dfrac{\frac{3}{2}t_0^2}{320} = 30 - t_0$

$t_0 = \dfrac{80}{3}$ sec

$h = \dfrac{3200}{3}$ m

$v = 3t_0 = 80$ m/s



5. A

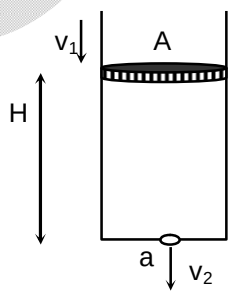
Sol. Height of water lands are 0.75 m and 0.25 m (for resonance)

$Av_2 = av_2$

$v_2 = \sqrt{2gH}$

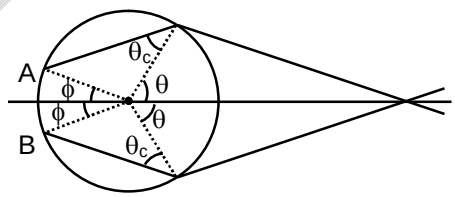
$a\sqrt{2gH} = A\left(-\dfrac{dH}{dt}\right)$

$\Rightarrow t \approx 66$ sec



6. A

Sol. Use concept of TIR



7. B

Sol. Smallest main scale division = 0.05 cm

Diamter = Main scale reading + Vernier coincidence L.C.

$= 5.10 + 24 \times 0.001 = 5.124$ cm

8. A

Sol. For central maxima optical path difference is zero.

9. C

Sol. Use Doppler's effect

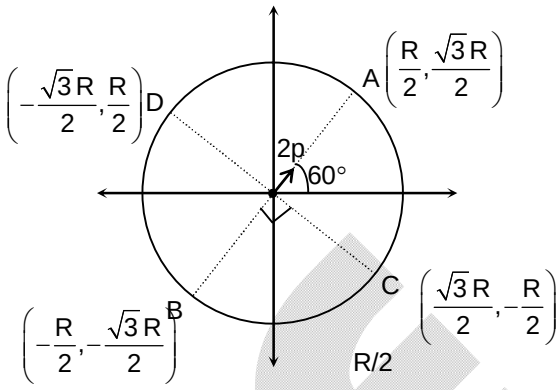
10. D

Sol. The resultant dipole moment has magnitude $\sqrt{(\sqrt{3}P)^2 + P^2} = 2P$ at an angle

$\theta = \tan^{-1}\dfrac{\sqrt{3}P}{P} = 60°$ with positive direction

Diameter AB is along net dipole moment and diameter CD is normal to net dipole momet.

$\therefore$ Potential at $A\left(\dfrac{R}{2}, \dfrac{\sqrt{3}R}{2}\right)$



Potential is zero at $C\left(\dfrac{\sqrt{3}R}{2}, -\dfrac{R}{2}\right)$ and $D\left(-\dfrac{\sqrt{3}R}{2}, \dfrac{R}{2}\right)$

Magnitude of electric field is $\dfrac{1}{4\pi\varepsilon_0}\dfrac{4p}{R^3}$ at $\left(\dfrac{R}{2}, \dfrac{\sqrt{3}R}{2}\right)$ and $B\left(-\dfrac{R}{2}, -\dfrac{\sqrt{3}R}{2}\right)$

Magnitude of electric field is $\dfrac{1}{4\pi\varepsilon_0}\dfrac{2p}{R^3}$ at $C\left(\dfrac{\sqrt{3}R}{2}, -\dfrac{R}{2}\right)$ and $D\left(-\dfrac{\sqrt{3}R}{2}, \dfrac{R}{2}\right)$

11. D

Sol. In photoelectric effect

$T = h\nu - \phi$

For A, T_A 4.25 $-\phi_A$

For B, T_B 4.70 $-\phi_B$

$T_B = T_A - 1.5$

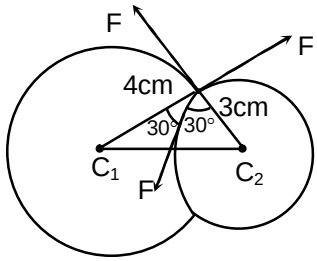
$\dfrac{\lambda_A}{\lambda_B} = \dfrac{h/\sqrt{2mT_A}}{h/\sqrt{2mT_B}} = \dfrac{1}{2} = \sqrt{\dfrac{T_B}{T_A}}$

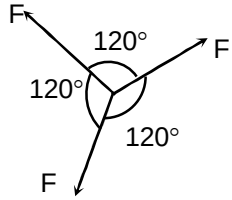
$T_A = 2eV$, $T_B = 0.5$ eV, $\phi_A = 2.25$ eV, $\phi_B = 4.20$ eV

SECTION – B

12. 13

Sol. $C_1C_2 = \sqrt{4^2 + 3^2 - 2\times 4\times 3\cos 60°} = \sqrt{13}$





13. 1

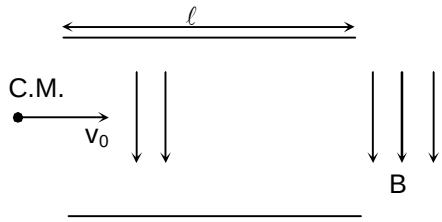
Sol. Use concept of ampere's law

14. 1

Sol. $\frac{1}{2}mv_0^2 = eV$

At instant t = t, acceleration of $e^- = \frac{eE}{M} = \frac{edt}{M}$

Pitch = $\frac{2\pi Ma\ell^2}{4eBv} = 9$ mm



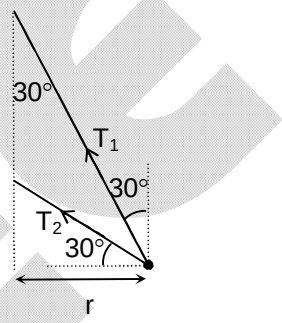
15. 1

Sol. $T_1\cos 30° + T_2\sin 30° = mg$

$T_1\sin 30° + T_2\cos 30° = \frac{mv^2}{r}$

$T_2 = -mg + \frac{\sqrt{3}mv^2}{r}$

$v_{min} = \sqrt{\frac{gr}{\sqrt{3}}} = 1$m/s

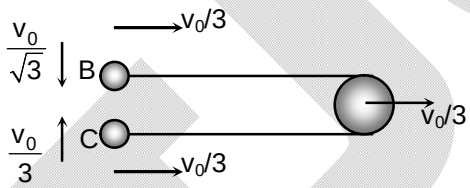


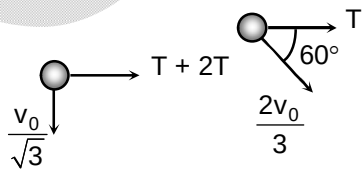
16. 16

Sol. $3T = \frac{mv_0^2}{3\ell}$

$T = \frac{mv_0^2}{9\ell}$

$r_C = \frac{4v_0^2 \times 18\ell}{9\times\sqrt{3}v_0^2} = \frac{8\ell}{\sqrt{3}}$





17. 2

Sol. $d\tau = \lambda R d\theta E R\sin\theta$

$\tau = \lambda_0 R^2 E\int_0^{2\pi}\sin^2\theta d\theta$

$\tau = \lambda_0 R^2 E\left[\left(\frac{1-\cos 2\theta}{2}\right)d\theta\right]_0^{2\pi}$

$\tau = \lambda_0\pi R^2 E$

$f_s = ma$ …(i)

$\lambda_0\pi R^2 E - f_s R = mR^2\alpha$

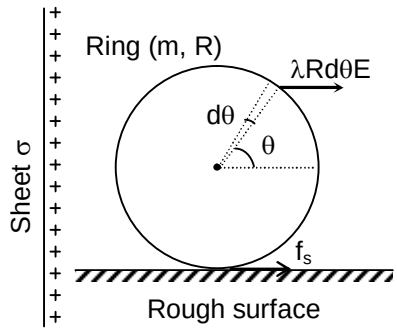
$\lambda_0\pi RE - f_s = ma$ …(ii)

Solving equations (i) and (ii), we get

$\lambda_0\pi RE = 2ma$

$a = \frac{\lambda_0\pi RE}{2m} = \frac{\lambda_0\pi R}{2m}\frac{\sigma}{2\varepsilon_0}$ $\left(\because\ E = \frac{\sigma}{2\varepsilon_0}\right)$

$a = \frac{\lambda_0\sigma\pi R}{4m\varepsilon_0}$



Chemistry

PART – II

SECTION – A

18. A, C

Sol. Balanced chemical equation

$$4Au + 8KCN + O_2 + 2H_2O \longrightarrow 4K\left[Au(CN)_2\right] + 4KOH$$

Weight of Au in 1 kg sample = 1.97 g

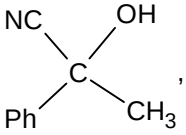
Moles $= \frac{1.97}{197} = 10^{-2}$

$\therefore$ Moles of KCN required $= 2 \times 10^{-2}$

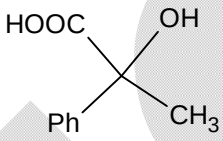
Weight of KCN required $= 2 \times 10^{-2} \times 65$

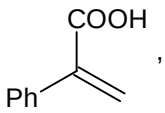
= 1.3 g.

19. A, B, C

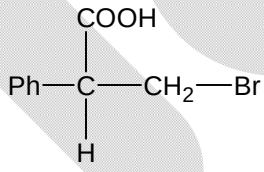
Sol. A : 

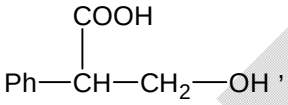
,

B = 

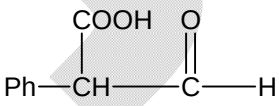
C : 

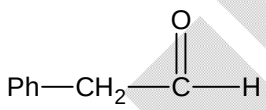
,

D = 

E : 

,

F = 

G : 

20. B, D

Sol. At anode

$$Ni(s) \longrightarrow Ni^{2+} + 2e^-$$

At cathode

$$Ni^{2+} + 2e^- \longrightarrow Ni(s)$$

$\therefore$ Mass of Ni electrode (cathode) increases.

$\therefore$ Number of equivalent of Ni^{2+} formed = number of equivalents of Ni^{2+} lost.

$\therefore$ No change in concentration of Ni^{2+} in the electrolytic solution.

21. B

Sol. I and III are correct.

22. B

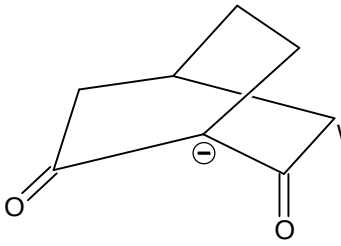
Sol. $\mu_{H_2O_2} > \mu_{H_2O}$

$O_2(g)$ – colourless $\quad O_2(\ell)$ – Blue

$$Ca(NO_3)_2 \xrightarrow{\Delta} CaO + NO_2 + O_2$$

23. D

Sol.



will not be stabilized by resonance.

24. B

Sol.

$$C_3H_7NH_2 + HCl \longrightarrow C_3H_7\overset{+}{N}H_3Cl^-$$

Intial	2mmol	2mmol	–
at equivalence point	0	0	2mmol

$$pH = 7 - \frac{1}{2}\left[pK_b + \log\left(\frac{2}{40}\right)\right]$$

$$\approx 6$$

25. B

Sol. $Cr^{+2}(HS) - d^4 = t_{2g}^3 e_g^1$

$Mn^{+3}(LS) - d^4 = t_{2g}^4 e_g^o$

$Fe^{+2}(HS) - d^6 = t_{2g}^4 e_g^2$

$Ni^{+4}(LS) - d^6 = t_{2g}^6 e_g^o$

26. D

27. C

Sol. $N_2O_5 + H_2O \longrightarrow HNO_3$

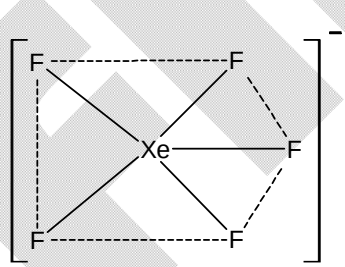
$N_2O_3 + H_2O \longrightarrow HNO_2$

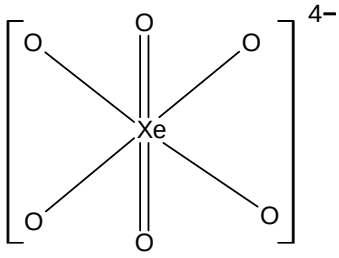
$Cl_2O_7 + H_2O \longrightarrow HClO_4$

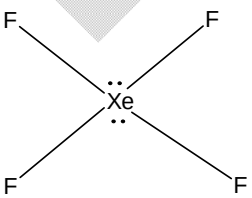
$Cl_2O + H_2O \longrightarrow HClO$

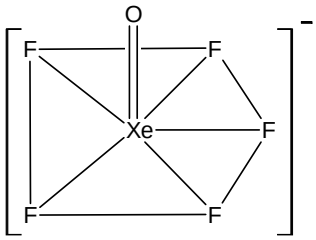
28. C

Sol.









SECTION – B

29. 9375

Sol. Energy required for dissociation of N_2 per molecule $= \dfrac{960\times10^3}{6\times10^{23}} = 16\times10^{-19}$ J / molecule

Since each N-atom has 2 eV extra energy after photochemical dissociation.

$\therefore$ Energy required for dissociation $= 16\times10^{-19} + 2\times1.6\times10^{-19}$

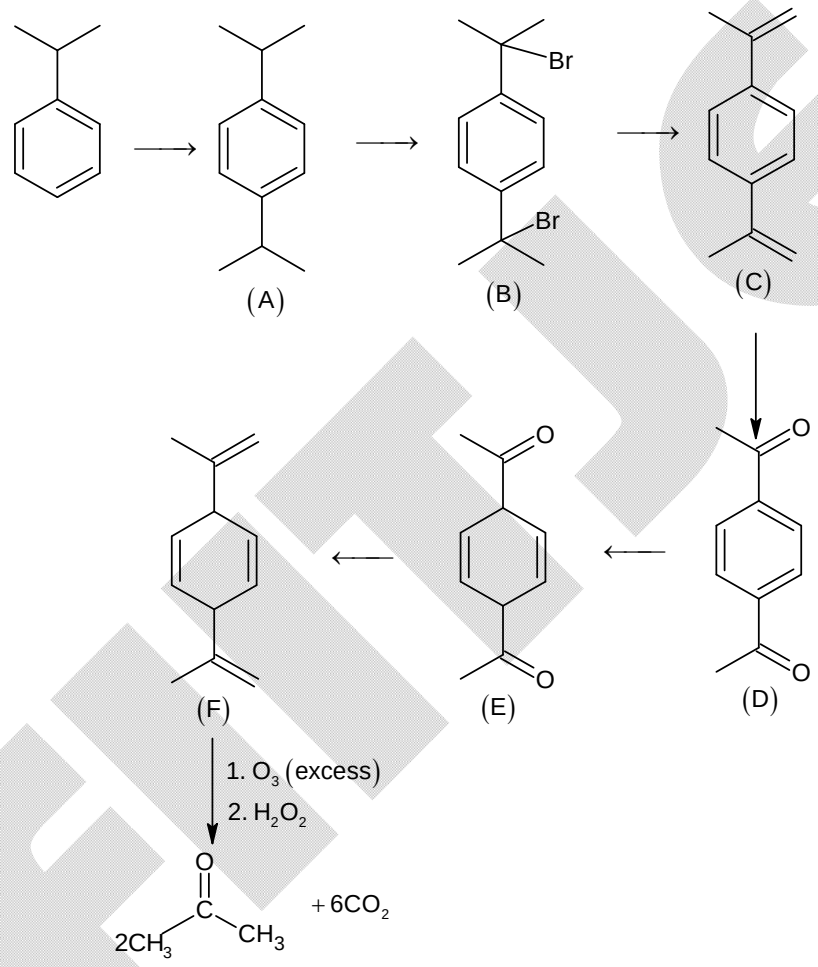
$= 19.2\times10^{-19}$ J

$\dfrac{hC}{\lambda} = 19.2\times10^{-19} \Rightarrow \lambda = 9375\times10^{-11}$ m.

$x = 9375$

30. 6

Sol.



31. 2000

Sol. $500 = \dfrac{1}{\lambda_{av}}\ell n\left(\dfrac{N_o}{N_t}\right)$

$500 = \dfrac{1}{\lambda_{av}}\ell n\left(\dfrac{N_o}{\frac{N_o}{8}}\right) = \dfrac{2}{\lambda_{av}}$

$\lambda_{av} = \dfrac{1}{250}$

$\lambda_{av} = \lambda_\alpha + \lambda_\beta = \dfrac{1}{t_\alpha} + \dfrac{1}{t_\beta}$

$\dfrac{1}{250} = \dfrac{1}{t_\alpha} + \dfrac{1}{400}$

$t_\alpha = \dfrac{2000}{3}$ years

$\therefore x = 2000$

32. 3

Sol. PbS, CuS, CuO – Black
Sb_3S_2 – Orange
CdS – Yellow
Cu_2O – Red

33. 1155

Sol. From the given data if is clear that reaction follows first order kinetics
$t_{1/2} = 50$ min
Partial pressure of X(g) remaining after 200 min

$(4 \times t_{1/2}) = 600 \times \left(\dfrac{1}{2}\right)^4 = 37.5$ mm Hg

	$2X(g)$	$\longrightarrow$	$2Y(g) + 3Z(g) + 4W(s)$	
Initial	600		0	0	–
$t = 200$ min	$600 - P = 37.5$		P	$\dfrac{3}{2}P$	

$\Rightarrow P = 562.5$
Total pressure after 200 min

$= 600 - P + P + \dfrac{3}{2}P$

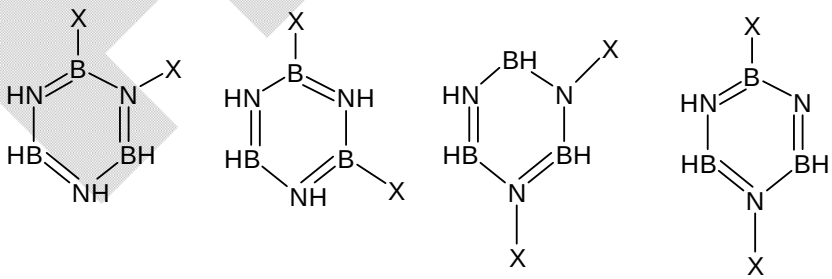
$= 600 + \dfrac{3}{2} \times 562.5$

= 1443.75 mm Hg.

$\dfrac{4}{5}P_{total} = 1155$

34. 3

Sol. $B_3N_3H_4X_2$ has 4-isomer.



Borazon is crystalline form of boron nitride.

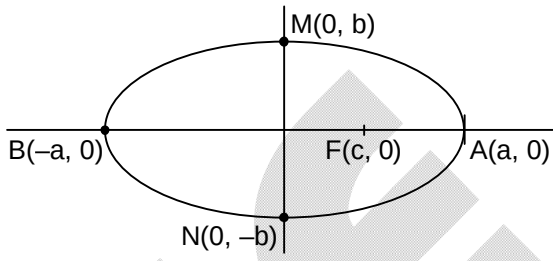
Mathematics

PART – III

SECTION – A

35. A, C, D

Sol. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ let $c^2=a^2-b^2 \Rightarrow c=\sqrt{a^2-b^2}$



FA = a – c $\Rightarrow$ FB = a + c $\Rightarrow$ FM = a = FN

Case-I:

If A, B are the left and figure vertices respectively

Then, a + c = 3, a – c = 2

|AB| = 2a = 5

According $b^2 = (a + e)(a – c) = 6$

So, then equation of ellipse is $\frac{4x^2}{25}+\frac{y^2}{6}=1$

Case-II:

Let A is left vertex and B is the top or bottom vertex, then a + c = 3, a = 2 and c = 1

Hence, $b^2 = a^2 – c^2 = 3$

So, $|AB|=\sqrt{a^2+b^2}=\sqrt{7}$ the corresponds of ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$

Case-III:

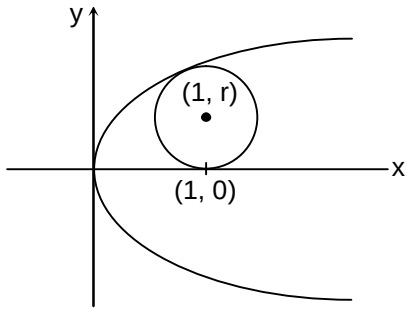
If A is the top or bottom vertex and B is the right vertex, then a = 3, a – c = 2, c = 1

$b^2 = a^2 – c^2 = 8$

$|AB|=\sqrt{a^2+b^2}=\sqrt{17}$

36. A, B

Sol. Equation of circle is $(x – 1)^2 + (y – r)^2 = r^2$



Put $x=\frac{y^2}{4}; \left(\frac{y^2}{4}-1\right)^2+(y-r)^2=r^2$

$\Rightarrow r=\frac{\left(y^2+4\right)^2}{32y}$

By the given condition equation has just one positive solution of which corresponds to the unique common point between circle and parabola. We shall consider the minimum value of

$f(y)=\frac{\left(y^2+4\right)^2}{32y}(y>0)$

Using AM·GM $y^2 + 4 = y^2+\frac{4}{3}+\frac{4}{3}+\frac{4}{3}\geq 4\sqrt{y^2\left(\frac{4}{3}\right)^3}$

$f(4)\geq\frac{1}{32y}\cdot 16\cdot\sqrt{y^2\left(\frac{4}{3}\right)^3}=\frac{4\sqrt{3}}{9}$

So, f(y) takes the minimum value $\frac{4\sqrt{3}}{9}$ if and only if $y^2=\frac{4}{3}$

$\Rightarrow y=\frac{2\sqrt{3}}{3}$ we get, $r\geq\frac{4\sqrt{3}}{9}$

So, $r = \frac{4\sqrt{3}}{9}$ is consistent with the given condition at this point $\left(\frac{1}{3}, \frac{2\sqrt{3}}{3}\right)$ is a unique common point between circle and parabola

37. A, B, C, D

Sol. f(x) is S.D. in (0, 3]
$\Rightarrow$ S.I. in [3, 9]
$\Rightarrow$ S.D. in [9, ∞) with f(3) = 0, f(9) = 1
$\Rightarrow a \in (0, 3), b \in (3, 9), c \in (9, \infty)$
Also, $f(a) = f(b) = f(c) \in (0, 1)$
As f(a) = f(b), then $1 - \log_3 a = \log_3 b - 1$
$\Rightarrow \log_3 a + \log_3 b = 2 \Rightarrow ab = 9$ and $abc = 9c$
Since, $0 < f(c) = 4 - \sqrt{c} < 1 \Rightarrow c \in (9, 16)$
$abc = 9c \in (81, 144)$

38. B

Sol. Use AM $\geq$ GM
$(1 + 2a)(2 + 2b)(2a + b) \geq 8\sqrt{2a \cdot 4b \cdot 2ab} = 32ab$
So, equally condition must be satisfied this means 2a = b = 1, so we have only one solution
Alternate solution:
$(1 + 2a)(2 + 2b)(2a + b) = 32ab \Rightarrow b(2a-1)^2 + 2a(b-1)^2 + (2a-b)^2 = 0$
Since, $a, b > 0 \Rightarrow 2a - 1 = b - 1 = 2a - b = 0 \Rightarrow (a, b) = \left(\frac{1}{2}, 1\right)$

39. D

Sol. The roots of $z^2 + z + 1 = 0$; $z = -\frac{1 \pm \sqrt{3}i}{2}$

So, $z^2 + z + 1 = \left(z - \left(\frac{-1+\sqrt{3}i}{2}\right)\right)\left(z - \left(\frac{-1-\sqrt{3}i}{2}\right)\right) \Rightarrow \left|z - \left(\frac{-1+\sqrt{3}i}{2}\right)\right| \cdot \left|z - \left(\frac{-1-\sqrt{3}i}{2}\right)\right| = 4$

Now let z = a + ib (we are trying to maximize b) $\Rightarrow \left|a + \frac{1}{2} + \left(b + \frac{\sqrt{3}}{2}\right)i\right| \left|a + \frac{1}{2} + \left(b - \frac{\sqrt{3}}{2}\right)i\right| = 4$

Since we are trying to maximize b, we want the real parts of the components to be as small as possible which we can do by setting $a = -\frac{1}{2}$

This leaves $\left|b + \frac{\sqrt{3}}{2}\right| \left|b - \frac{\sqrt{3}}{2}\right| = 4 \Rightarrow b^2 - \frac{3}{4} = 4 \Rightarrow b^2 = \frac{19}{4} \Rightarrow b = \frac{\sqrt{19}}{2}$

40. B

Sol. These are $10^2 = 100$ equally likely selection for array (a, b), let us consider the number of selections say N that $a^2 + b$ is divisible by 3
If a is divisible by 3, then b is also divisible by 3. In this case, there are 3 selection for a, b respectively, then the number of selection for (a, b) is $3^2 = 9$
If a is not divisible by 3, then $a^2 - 1$ is divisible by 3 and hence b + 1 is also divisible by 3, in this case a has 7 selection and b has 4. So, that 28 outcomes probability = $\frac{37}{100}$

41. D

Sol. $\frac{x^2}{a^2} + \frac{y^2}{b^1} = 1$ (a > b > 0)

$|AB| = 2a = 4,\ a + \sqrt{a^2 - b^2} = |AF| = 2 + \sqrt{3} \Rightarrow a = 2,\ b = 1,\ |EF| = 2\sqrt{a^2 - b^2} = 2\sqrt{3}$
$\because\ |PE| + |PF| = 2a = 4$ and $|PE| \cdot |PF| = 2$
Then $|PE|^2 + |PF|^2 = (|PE| + |PF|)^2 - 2|PE| \cdot |PF| = 12 = |EF|^2$
i.e., $\angle EPF$ is right angle $\Delta PEF = \frac{1}{2}|PE| \cdot |PF| = 1$

42. D
Sol. Total number is $4 \times 4! = 96$
Permutation (2, 0, 1, 9, 2019)(2019, 2, 0, 1, 9)
Correspondence to the same number 20, 192019, while other are different
So, 96 – 1 = 95

43. B
Sol. (P) Let $x = \sin\theta$

$$\int_0^{\pi/2} \frac{\sin^2\theta \cdot \ln(\sin\theta)(\cos\theta)}{\cos\theta} d\theta \Rightarrow \int_0^{\pi/2} \left(\frac{1-\cos 2\theta}{2}\right) \ln(\sin\theta) d\theta \Rightarrow \frac{\pi}{8}(1 - \ln 4)$$

(Q) $$\int_0^{\pi} \frac{-\sin^{-1}\{x\}}{\sqrt{1-(\{x\})^2}} dx \Rightarrow -10\int_0^1 \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx = -10\left(\frac{\pi^2}{8}\right) = -\frac{5\pi^2}{4}$$

(R) $$\int_{-3}^{3} f(x)dx = -\frac{3}{2}$$

(S) $$\int_0^{20} \sqrt{2}\left|\sin\frac{\pi x}{2}\right| dx = 10\sqrt{2}\int_0^2 \sin\left(\frac{\pi x}{2}\right) dx$$

44. D
Sol. (P) f is differential in [0, 5] f(0) = 4, f(5) = –1

$$\Rightarrow g(x) = \frac{f(x)}{x+1},\ g(0) = \frac{4}{1} = 4$$

$$\Rightarrow g(5) = -\frac{1}{6}$$

By LMVT $g'(c) = \frac{g(5) - g(0)}{5 - 0} = -\frac{5}{6}$

(Q) By LMVT

$$f'(c) = \frac{f(5) - f(1)}{5 - 1} = \frac{0 - \sqrt{24}}{4}$$

$$\Rightarrow \frac{-c}{\sqrt{25 - c^2}} = \frac{-\sqrt{24}}{4}$$

(R) By LMVT on [1, 6]

$$f'(c) = \frac{f(6) - f(1)}{6 - 1}$$

$\Rightarrow f'(c) \geq 2 \Rightarrow f(6) \geq 8$

(S) f(x) and g(x) are differentiable for $x \in [0, 1]$, f(0) = 2, g(0) = 0, f(1) = 6
$\because\ f'(c) - 2g'(c) = 0$ for some $c \in (0, 1)$
$\Rightarrow f(0) - 2g(0) = f(1) - 2g(1)$
$\Rightarrow 2 - 2 \times 0 = 6 - 2g(1)$
$\Rightarrow g(1) = 2$

45. B

Sol. (P) Use ${}^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r}$

SECTION – B

46. 8321

Sol. $\prod_{k=0}^{12}(2-2\omega^{k}+\omega^{2k}) = \prod_{k=0}^{12}\left((1-\omega^{k})^{2}+1\right)$

$\Rightarrow \prod_{k=0}^{12}((1+i)-\omega^{k})((1-i)-\omega^{k})$

We consider the polynomial $x^{13}-1=0$ whose roots are the 13th root of unity

Let $\omega^{0}, \omega^{1}, \omega^{2}, ..., \omega^{12}$ are the roots of this equation

$\Rightarrow x^{13}-1=(x-\omega^{0})(x-\omega^{1})(x-\omega^{2})(x-\omega^{3}).....(x-\omega^{12})$

Put x = 1 + i and 1 – i

$\Rightarrow \prod_{k=0}^{12}((1+i)-\omega^{k})((1-i)-\omega^{k}) \Rightarrow \left(\sqrt{2}e^{i\frac{\pi}{4}}-\omega^{k}\right)\left(\sqrt{2}\cdot e^{-i\frac{\pi}{4}}-\omega^{k}\right)$

$\Rightarrow \left(\left(\sqrt{2}\cdot e^{i\frac{\pi}{4}}\right)^{13}-1\right)\left(\left(\sqrt{2}\cdot e^{-i\frac{\pi}{4}}\right)^{13}-1\right) \Rightarrow \left(2^{\frac{13}{2}}\cdot e^{i\frac{13\pi}{4}}-1\right)\left(2^{\frac{13}{2}}\cdot e^{-i\frac{13\pi}{4}}-1\right)$

$\Rightarrow 2^{13}+1+2^{\frac{13}{2}}\cdot 2\cos\frac{\pi}{4} = 2^{13}+1+2^{7} = 8321$

47. 18

Sol. If we use the product rule to calculate $f_{n}''(x)$, the result is a sum of terms of two types, terms where two distinct factors $\cos(m_1 x)$ and $\cos(m_2 x)$ have each been differentiated once, and terms where a single factors cos(mx) has been differentiated twice. When we calculate at x = 0, all terms of first type vanish since sin(0) = 0, while term of the second type involving (cos mx)″ becomes $(-m^2)$, thus

$$\left|f_{n}''(0)\right| = \left|-\sum_{m=1}^{n}m^{2}\right| = \frac{n(n+1)(2n+1)}{6}$$

The function $g(n)=\frac{n(n+1)(2n+1)}{6}$ is increasing for $n \in N$ and satisfy g(17) = 1785 and g(18) = 2109

So, n = 18

48. 1

Sol. For $\log_{7x}2023\cdot\log_{289x}2023 = \log_{2023x}2023$

$\Rightarrow \log_{2023}7x\cdot\log_{2023}289x = \log_{2023}2023x$

Let $\log_{2023}x = a$

We get, $(\log_{2023}7+a)(\log_{2023}289+a) = 1+a$

$\Rightarrow a^{2}+(\log_{2023}7+\log_{2023}289)a+\log_{2023}7\cdot\log_{2023}289 = 1+a$

$\Rightarrow a^{2}+\log_{2023}7\cdot\log_{2023}289-1=0 \Rightarrow a_1 + a_2 = 0$

$\log_{2023}x_1+\log_{2023}x_2 = 0 \Rightarrow x_1\cdot x_2 = 1$

49. 5

Sol. Let the common difference = d

We know that $a_1 + a_2 = a_k$ (k is some positive integer)

$\Rightarrow 2a_1 + d = a_1 + (k-1)d$ i.e. $a_1 = (k-2)d$

$\Rightarrow d = \dfrac{a_1}{k-2}, k \neq 2$

Hence, $a_n = a_1 + (n-1)d = a_1 + \dfrac{n-1}{k-2} \cdot a_1$

So, for any positive integer n_1

$a_1 + a_2 + ..., a_n = a_1 \cdot n + \dfrac{n(n-1)}{2} \cdot d = a_1 + \left((n-1)(k-2) + \dfrac{n(n-1)}{2}\right)d$

The above is really a term in $\{a_n\}$ therefore, we only need to consider the number of positive integer k such that $\dfrac{a_1}{(k-2)}$ is integer

Note that 2019 is a product of primes 3 and 673, then it is easy to find that the values of k – 2 are –1, 1, 3, 673, 2019 which corresponds to 5 AP

50. 4

Sol. We have equation of circle $(x - (r+1))^2 + y^2 = r^2$

$\Rightarrow (a - r - 1)^2 + b^2 = r^2 \Rightarrow r^2 - (a - r - 1)^2 = b^2$

$(r + a - r - 1)(r - a + r + 1) = b^2 \Rightarrow (a-1)(2r - (a-1)) = b^2$

$2r(a-1) - (a-1)^2 = b^2 \geq 4a \Rightarrow 2ra - 2r - (a^2 - 2a + 1) - 4a \geq 0$

$2ra - 2r - a^2 + 2a - 1 - 4a \geq 0 \Rightarrow -a^2 - 2a - 2r + 2ar - 1 \geq 0$

$a^2 + 2a + 2r - 2ar + 1 \leq 0 \Rightarrow a^2 - 2a(r-1) + 2r + 1 \leq 0$

This equation has solution $(2(r-1))^2 - 4(2r+1) \geq 0$

$4r(r-4) \geq 0 \Rightarrow r \geq 4$, (a, b) = $\left(3, 2\sqrt{3}\right)$ satisfy the given condition when $r \geq 4$

Minimum value of r = 4

51. 51

Sol. $\left(x + 2\sqrt{y} - 1\right)^n = \sum_{r=0}^{n} {}^nC_r \cdot x^{n-r} \cdot \left(2\sqrt{y} - 1\right)^r$

So, coefficient of x^{n-4} is ${}^nC_4 \cdot x^{n-4} \cdot \left(2\sqrt{y} - 1\right)^4$

$(-1)^4 \cdot {}^nC_4 = \dfrac{n(n-1)(n-2)(n-3)}{24}$ ..... (1)

Coefficient of xy, ${}^nC_{n-1} x \left(2\sqrt{y} - 1\right)^{n-1}$

${}^nC_{n-1} \cdot {}^{n-1}C_2 4(-1)^{n-3} = (-1)^{n-3} 2n(n-1)(n-2)$ ..... (2)

From equation (1) and (2), we get n = 51